

VAMSHI KRISHNA

Data Scientist

Email: connects2vk@gmail.com | 312-291-1294 | [LinkedIn](#)

Professional Summary

- Experienced Senior Data Scientist having 10+ years of experience in the Healthcare, Financial, Telecom, Banking and Retail Domains using predictive modeling, data processing, data mining, and Forecasting algorithms to solve challenging business problems.
- Expertise in Python, R, and SQL. Experience in using various packages in Python like Pandas, NumPy, PySpark, MLlib, SciPy, Matplotlib, Seaborn, TensorFlow, Scikit-Learn, ggplot2, PyTorch, PyODBC, PyPDF2, NLTK, Keras, Flask, Shutil, CxOracle, win32com.client, extract_msg,
- Proficient in R for data analysis and statistical modeling. Experience in using packages like Data.table, dplyr, arules, stringr, plyr, zoo, tidyr, statsmodels, tseries, DIB, snowflake, forecast, ggplot2, survival, caTools, lpsolve, caret, cvms, ConfusionTableR.
- Strong hands-on experience in Data Collection, Data Cleaning, Exploratory Data Analysis(EDA), Data Mining, Data wrangling, Data modeling, Statistical Modeling, Feature Engineering, Data Migration, Transformation, Integration Survival Analysis, and Linear Optimization.
- Strong Hands-on Experience in Machine Learning algorithms and Predictive Modeling including Regression, Clustering, Tree based algorithms(Decision Tree, Random Forest, ID3, AdaBoost, XGBoost), SVM, Naïve Bayes, K-Means, Neural Networks (RNN, CNN, LSTM), Time Series Forecast (ARIMA), Apriori.
- Developed an LLM model using GPT, BERT, and RoBERTa to analyze the sentiment of customers.
- Developed a Sentiment analysis model using Natural Language Processing (NLP) to categorize email. Utilized a sentiment lexicon to assign sentiment scores to words and determine the overall sentiment and the context of the email.
- Designed a Long Short-Term Memory (LSTM) model for sentiment analysis, achieving higher accuracy compared to traditional machine learning algorithms.
- Successfully applied data transformation techniques such as data smoothing (clustering, binning), normalization, integration, blending, and feature engineering to enhance data quality and facilitate analysis in previous projects.
- Utilized clustering algorithms to group similar data points, developed normalization techniques to standardize data, and performed feature engineering to create new variables that improved model performance.
- Well versed with using resampling methods such as Oversampling (SMOTE) and Under sampling (Cluster Centroid) techniques for balancing the datasets.
- Experienced at creating data regression models, using predictive data modeling, and analyzing data mining algorithms to deliver insights and implement action-oriented solutions to complex business problems
- Experience with Artificial Neural Networks: Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN).
- Experience with Supervised Machine Learning algorithms like Logistic Regression, Linear regression, K- Nearest Neighbors, Naive Bayes, Support Vector Machines, Decision Trees and Ensemble models: Random Forests, Gradient Boosted Decision Trees.
- Experience with Unsupervised Machine Learning algorithms like K-Means, Density Based Clustering (DBSCAN), Hierarchical Clustering (Agglomerative, Divisive) and good knowledge on Recommender Systems.
- Experience using Dimensionality Reduction Techniques like Principal Component Analysis, and Linear Discriminant Analysis (LDA).
- Extensive knowledge with Statistical methodologies such as Hypothesis Testing, and Time Series Analysis.
- Capable of creating complete NLP pipeline with machine learning, feature creation and union, and parameter tuning.
- Experience with AWS ecosystem and Implemented the best BI practices in Microsoft Power BI and Tableau.
- Have the experience of creating time series graphs, Geo graphs, Pivot charts, box plots, donuts in power bi desktop and Tableau desktop.
- Solid ability to write and optimize diverse SQL queries, working knowledge of RDBMS like SQL Server, Oracle, Google Big query, MySQL.
- Excellent communicative, interpersonal, intuitive, analysis and leadership skills with ability to work efficiently in both independent and teamwork environments.

TECHNICAL SKILLS

- **Cloud Platforms:** AWS (Bedrock, SageMaker, OpenSearch Service, ECS Fargate, Lambda, Textract, Glue, Athena, S3, EMR, Glue Crawlers, Lake Formation, CodePipeline, CloudWatch, API Gateway, VPC, IAM), Google Cloud (Vertex AI, PaLM API, BigQuery, GKE, Cloud Storage, Cloud Composer, Document AI, Cloud Vision API), Azure (Machine Learning, Data Factory, Synapse, HDInsight, Blob Storage, Cosmos DB, Databricks, AKS)
- **Programming & Data Analysis:** Python (Pandas, NumPy, PySpark, PyTorch MLib, SciPy, Matplotlib, Seaborn, TensorFlow, Scikit-Learn, Keras, PyODBC, PyPDF2, NLTK, Flask, Shutil, CxOracle, win32com.client, extract_msg), R (Data.table, dplyr, arules, stringr, plyr, zoo, tidyr, statsmodels, tseries, DIB, snowflake, forecast, ggplot2, survival, caTools, lpsolve, caret, cvms, ConfusionTableR), SQL (SQL Server, Oracle, MySQL, Google BigQuery), Unix Shell Scripting
- **ML Libraries & AI Tooling:** LangChain, LangGraph, Amazon Bedrock, Retrieval-Augmented Generation (RAG), Amazon OpenSearch k-NN, MLflow, SageMaker Pipelines, Vertex AI Pipelines, Llama 3, FAISS, PaLM API, Model Monitor, LlamaIndex, Hugging Face Transformers, XGBoost, LightGBM, CNNs, Autoencoders, Isolation Forest, SHAP, LIME, Statsmodels, spaCy, NLTK, OpenCV, Tesseract OCR
- **Data Warehousing & Storage:** Amazon OpenSearch Service, Redshift, BigQuery, Snowflake, Azure Synapse, Teradata (FastLoad, MultiLoad, TPT), HDFS, Oracle, SQL Server, MySQL, PostgreSQL, Cassandra, HBase
- **Visualization & Reporting:** Tableau, Power BI, Looker, SSRS, Excel, Power Query
- **DevOps & Version Control:** Docker, Kubernetes (GKE, AKS), Jenkins, Git, GitHub, GitLab, CI/CD, Blue/Green Deployments, Confluence, SharePoint

Professional Experience

Client: CVS Health, Woonsocket, RI

Oct 2023 - Present

Sr Data Scientist/ GenAI Engineer Responsibilities:

- Led Agile-driven healthcare data initiatives, participating in Scrum ceremonies, sprint planning, and iterative development cycles to enhance data processing solutions.
- Revised and maintained scalable, HIPAA-compliant healthcare data pipelines using Hadoop, Spark, Kafka, and AWS Glue, enabling real-time EHR data processing with 26M users managing over 50 TB of patient and operational data on AWS platform
- Built and automated data pipelines integrating claims, EMR/EHR, and pharmacy data, ensuring scalable analytics for healthcare utilization and cost analysis.
- Consolidated patient records, claims data, and operational datasets from various sources using FHIR-based REST APIs, ensuring seamless updates and improving dashboard freshness for executive use by automating real-time data updates, reducing manual intervention by 40%
- Automated data validation and transformation in Salesforce and EHR Systems using Python and Apache Spark, reducing errors by 30% and improving reporting accuracy.
- Developed, tested, and maintained Python-based automation solutions, including Salesforce and EHR automation scripts using Selenium, reducing manual interventions and increasing efficiency
- Developed HIPAA-compliant workflows for clinical data storage and analytics on AWS and Salesforce, achieving 100% adherence to regulatory standards achieving a 30% improvement in prediction accuracy for patient readmission risk
- Implemented and customized Salesforce-based applications using Apex and Visualforce, improving CRM workflows and customer satisfaction scores by 18%. Automated data integration processes to support better client management and operational efficiency.
- Developed automation scripts for EHR data extraction, validation, and transformation in healthcare systems, leveraging Selenium for web automation and BeautifulSoup for web scraping to streamline data ingestion
- Implemented Jenkins-based CI/CD pipelines for automated deployments, reducing manual effort by 40% and ensuring zero downtime during updates
- Built and maintained robust data pipelines for a healthcare client using Python and SQL, supporting the extraction, transformation, and loading (ETL) of large-scale data.
- Collaborated with cybersecurity teams to deploy anomaly detection frameworks for EHR access logs, decreasing incident response times by 15% and strengthening system resilience and data security
- Leveraged **Generative AI and LLMs** to enhance customer service chatbots, medication FAQs, and member engagement platforms, improving response accuracy and patient satisfaction.
- Designed **A/B testing frameworks** to measure the impact of AI-driven interventions on patient outcomes, medication adherence, and program effectiveness.

- Collaborated with data engineering teams to integrate **multi-modal healthcare data sources** (EHR, claims, pharmacy, IoT/wearables), creating enriched datasets for advanced ML modeling.
- Designed Power BI and Salesforce dashboards to track patient care trends, provider performance, and claims processing, increasing operational efficiency by 20% and enabling better resource allocation.
- Conducted root-cause analyses for system failures and compliance breaches, developing mitigation strategies that reduced recurring incidents by 20%, ensuring uninterrupted healthcare data operations.
- Seamlessly incorporated APIs to manage end-to-end data communication in Generative AI applications.
- Designed, developed, and launched chatbots and virtual assistants leveraging RAG methodologies, boosting customer engagement and operational efficiency.
- Applied NLP techniques to unstructured clinical notes and call center transcripts, improving patient engagement strategies and supporting personalized care recommendations.
- Enhanced natural language processing (NLP) solutions, improving sentiment analysis, named entity recognition (NER), text classification, and topic modeling.
- Demonstrated expertise in deploying and optimizing models using TensorFlow, PyTorch, and Hugging Face Transformers, achieving superior performance in AI projects.
- Optimized chatbot performance using LangChain and large language models (LLMs) by integrating diverse query engines.
- Expertly applied LLM architectures, including Transformers, BERT, and BART, achieving significant improvements in training efficiency and model output quality.
- Enhanced productivity through RAG-powered solutions for summarization and chatbot functionalities, leveraging LangChain and retrieval techniques.
- Developed robust RAG-based models using diverse vector databases for scalable data handling.
- Leveraged LLMs and RAG in application development to enhance user experiences with LangChain and OpenAI technologies. Developed and scaled machine learning and deep learning models like Logistic Regression, Random Forest, Gradient Boosting Machines, SVM (Support Vector Machines), etc. for classification
- Programmed using python to prototype and deploy Machine Learning, Deep Learning, Predictive models, Probabilistic and Statistical Modeling based approach with user interface development
- Build, train and deploy ML models on AWS with Amazon SageMaker, Used machine learning modelling techniques like GridSearchCV, SMOTE.
- Perform unit testing and provide system test support and Validate & monitor deliverable in production
- Used PuTTY for accessing the AWS EC2 instances for training the models with Production phase data
- Integrated the AWS server with GitLab to push the model into production and to monitor the performance
- Use Jira for project management teams such as report, & analysis, workflow customization, issue /task management, project customization and help to team of all types manage work
- Performed Exploratory Data analysis (EDA) to further explore the dataset, detect the outliers and extract important variables both graphically and numerically. Performed feature scaling (Standardization/standard scaling) and Min. Max scaling (Normalization).
- Used hyper parameter tuning methods like Bayesian optimization, random search and grid search for model tuning.

Client: Fannie Mae, Reston, VA

Nov 2021 to Sep 2023

Sr.Data Scientist Responsibilities:

- Developed and deployed **predictive credit risk models** using Python, R, and ML frameworks to assess borrower default probability, prepayment risk, and loan performance across Fannie Mae's mortgage portfolio.
- Developed various Clustering algorithms for market segmentation to analyze the customer behavior patterns.
- Built **machine learning pipelines** (XGBoost, Random Forest, Gradient Boosting, Neural Networks) to enhance borrower creditworthiness evaluation, reducing false positives in risk classifications.
- Applied **NLP and text analytics** on loan documents, appraisal notes, and servicing data to detect anomalies, improve underwriting accuracy, and flag potential fraud.
- Performed NLP by using some techniques like Word2Vec, BOW (Bag of Words), tf-idf, Avg-Word2Vec, if- idf, Weighted Word2Vec.
- Performed Sentimental analysis (NLP) on the email feedback of the customers to determine the emotional tone behind the series of words and gain the express of the attitudes and emotions by Long-Short Term Memory (LSTM) cells in Recurrent Neural Networks (RNN).
- Created **loan-level forecasting models** to project delinquency rates, loss severity, and prepayment speeds under varying macroeconomic scenarios, supporting stress-testing and capital planning.

- Leveraged **cloud platforms (AWS)** to scale ML training on millions of loan records, improving model runtime and accuracy for portfolio analytics.
- Developed **simulation models** to evaluate loan modifications, forbearance impacts, and repayment strategies during economic downturns, informing Fannie Mae's policy and servicing strategies.
- Created **interactive dashboards and visualizations** (Tableau, Power BI) for senior executives to monitor loan performance, credit risk trends, and predictive model outputs.
- Built machine learning models and constructed multilayer perceptions for Deep Neural Networks (DNN) to identify fraudulent applications for loan pre-approvals and to identify fraudulent credit card transactions by comparing the previous transactions.
- Implemented algorithms such as Principal Component Analysis (PCA) for dimensionality reduction and to normalize the large datasets.
- Used Pandas, NumPy for Data cleaning, (csv files clean), Matplotlib, Seaborn (heat map and visualization), Scikit-learn (Traditional ML models like), NLTK in Python at various stages for developing machine learning model.
- Participated in features engineering such as feature generating, PCA, feature normalization and label encoding with Scikitlearn preprocessing.
- Used cross-validation techniques to avoid the overfitting of the model to make sure the predictions are accurate and also measure the performance using Confusion matrix and Classification report.
- Used Conditional Random Fields for Parts of Speech tagging to recognize the similarities and differences between words.
- Implemented LSTM layer network of moderate depth to gain the information in the sequence with help of Tensor Flow.
- Performed topic modeling using Linear Discriminant Analysis (LDA) to identify the appropriate conversation to its service. Used Genism to perform Word2Vec and used LSTM cells to perform sentiment analysis.
- Created distributed environment of Tensor Flow across multiple devices (CPUs and GPUs).
- Used PySpark Machine learning library to build and evaluate different models.
- Built Multi-Linear classification algorithm for performing real-time text mining scoring on extracted data.
- Built time series model for complex pattern recognition with exponential smoothening of financial and multivariate time series data for short-term forecasts and forecast of returns.
- Used Computer Vision along with ResNet CNN and t-sne for image-based clustering and training neural network models to classify certain type of documents based on images.
- Used AWS Extract and AWS comprehend to extract data and analyze the meaning and relationships in text to identify key entities using tokenization and automatically organizes a collection of text files by topic.
- Performed EDA analysis and data mining (cleaning & manipulation) to create training and testing data sets for machine learning predictive models.
- Created AWS lambda functions for several micro services that runs behind the virtual assistant with ability to run services automatically using AWS CloudWatch triggers with cron expressions or when an API is called.
- Utilized data visualization tools such as Tableau and Python's vast data visualization libraries to communicate findings to the data science, marketing and engineering teams.
- Built complex ensemble machine learning algorithms using Amazon Sage Maker with automated training, testing and deployment of the ML models into production that can run at specific intervals using AWS Lambda.
- Created number of AWS lambda functions with layers which gives the ability to edit the code inline and automatic code runs using AWS CloudWatch triggers with different cron expressions.
- Involved in building CI/CD pipelines to automate the deployments using docker and Jenkins which is triggered automatically if there is a change in GitHub master branch, then prepare packages and deploy in Dev, QA environments for unit testing, static and composite analysis of the code and deploy to Production.

Client: Target, Minneapolis, MN

July 2020 – Sept 2021

Data Scientist Responsibilities:

- Implemented real-time recommendation systems for online shoppers, boosting conversion rates with collaborative filtering and deep learning algorithms.
- Developed machine learning models to optimize inventory levels and demand forecasting, improving stock availability and reducing overstock.
- Built predictive analytics solutions to segment customers and personalize marketing campaigns, resulting in a increase in key engagement metrics and ROI.
- Led cross-functional teams to deploy AI-driven pricing and promotions models, increasing gross margin by 15% in seasonal campaigns.

- Automated analysis of transactional and behavioral retail data using Python and TensorFlow, improving operational efficiency and accelerating insights delivery by 30%.
- Designed and monitored ML pipelines in cloud environments (Azure), enabling scalable deployment and reliability for high-volume retail data feeds.
- Conducted A/B testing and cohort analysis for loyalty programs, driving a 10% improvement in customer retention and lifetime value.
- Presented actionable insights and interactive dashboards to stakeholders using Tableau and Power BI, guiding strategic decisions in merchandising and customer experience.
- Enhanced fraud detection capabilities in retail e-commerce workflows using anomaly detection and supervised ML techniques, reducing chargebacks by 12%.
- Collaborated with product, engineering, and data teams to translate business requirements into robust AI solutions tailored for retail operations.

Client: AT&T, Plano Texas

Jan 2019 to June 2020

Machine Learning Engineer Responsibilities:

- Worked with stakeholders to gather requirements for churn prediction and customer segmentation, aligning business objectives with data science solutions.
- Collected, cleansed, and transformed large-scale telecom customer data using Python and SQL for model readiness.
- Engineered advanced features including call patterns, usage behavior, and billing anomalies to improve model performance.
- Built and tested classification models (Logistic Regression, Random Forest, Gradient Boosting) to predict high-risk churn customers.
- Designed clustering models (K-Means, DBSCAN) to create behavior-driven customer segments for targeted campaigns.
- Deployed ML models on Azure Machine Learning for real-time churn scoring and campaign recommendation engines.
- Automated data ingestion and scoring pipelines using Azure Data Factory and Python scripts for continuous model updates.
- Developed Power BI dashboards visualizing churn risk, revenue leakage, and campaign effectiveness for business teams.
- Performed model monitoring and retraining to handle concept drift and maintain predictive accuracy over time.
- Authored technical documentation and conducted workshops for marketing and product teams to interpret AI-driven insights effectively.
- Spearheaded cloud-native ETL processes, using Python APIs and Azure Data Factory for seamless data migration and integration.
- Designed and optimized data pipeline architectures in Microsoft Azure, enhancing performance and scalability with Data Factory and Databricks.
- Proficient in Python and PySpark, executed complex data transformations and maintained data integrity during migration to Azure Blob storage.
- Implemented cutting-edge Generative AI techniques in Azure Cognitive Search, enabling sophisticated similarity searches on PDF documents.
- Automated daily Health check reports delivery to stakeholders using Logic App solutions, ensuring timely insights.
- Ensured optimal operational efficiency with robust Azure Monitoring Service solutions, tracking daily job execution and performance.
- Utilized a diverse toolkit including Snowflake, Databricks, and Azure Data Factory for seamless data integration and transformation.
- Implemented advanced encryption methodologies, ensuring data security and compliance using AES encryption and Key Vault. Leveraged Azure Cognitive AI capabilities to enhance data discovery and retrieval processes.
- Utilized Databricks notebooks for interactive analysis, deriving actionable insights using Spark APIs.
- Ensured data quality, consistency, integrity using Pandas, NumPy.
- Conducted feasibility analysis and defined solution architecture for predictive analytics pipelines leveraging Azure Machine Learning services.
- Engineered end-to-end data pipelines using Python, PySpark, and Azure Data Factory for ingestion, transformation, and feature engineering.
- Designed and developed supervised and unsupervised ML models (regression, classification, clustering) to drive actionable insights for business units.
- Built deep learning models using TensorFlow and PyTorch for customer churn prediction and personalized recommendation engines.

- Applied advanced time-series modeling techniques (ARIMA, Prophet, LSTM) to improve accuracy of demand forecasting.
- Implemented hyperparameter tuning and model optimization using Azure ML HyperDrive and cross-validation strategies.
- Developed reusable Python modules and ML utilities for data preprocessing, anomaly detection, and model explainability (SHAP, LIME).
- Deployed ML and deep learning models as REST APIs on Azure Kubernetes Service (AKS) ensuring scalability and lowlatency inference.
- Integrated Azure Cognitive Services for NLP-based sentiment analysis and text classification in customer feedback analysis.
- Balanced classes in large data sets using resampling methods like Synthetic Minority Over Sampling Technique (SMOTE).
- Established CI/CD pipelines with Azure DevOps for automated model retraining, deployment, and monitoring in production environments.
- Designed data visualization dashboards using Power BI and Azure Synapse Analytics to enable interactive exploration of model outputs.
- Collaborated with data engineering teams to ensure model performance optimization with distributed computing and GPU-accelerated training.
- Worked closely with internal stakeholders such as business teams, product managers, engineering teams, and other teams.

Client: Citi Bank, Pune, India

Aug 2016 to Oct 2018

Python/ ETL Developer Responsibilities:

- Worked with stakeholders to gather requirements for churn prediction and customer segmentation, aligning business objectives with data science solutions.
- Experience working with Python, focusing on developing scalable applications, automation scripts, and data-driven solutions.
- Automated repetitive tasks and processes using Python scripts, significantly reducing manual effort and improving operational efficiency.
- Utilized Python libraries such as Pandas, NumPy, and Matplotlib for data manipulation, visualization, and analysis, driving better decision-making.
- Worked on large volumes of data and complex ETL Transformations by writing the complex SQL queries.
- Built python pipelines to query, load, and copy data. Experience with creation and maintenance of data pipelines for data ingestion.
- Experience with Extraction, Transformation and Loading of data from Multiple Sources to Data Warehouse.
- Completed required data profiling for DataMarts and documented all the steps from development to coding.
- Compared and checked the data models to adhere the given appropriations.
- Analyzed data and completed the process of integration of SQL Server Integration Services (SSIS).
- Used various transformations in SSIS Dataflow, Control Flow using For-loop containers and Fuzzy Lookups.
- Performed ETL from Oracle database using SAS and SQL to create datasets, Macros and formats according to the given requirement.
- Performed conditional split, derived column, lookup during data scrubbing, checking data validations for different stages.
- Validated reports using SQL queries on Oracle database.
- Created SAS Datasets, performed data manipulation, wrote test reports and tested data integrity and data validation.
- Examine customer feedback and activity for use in detecting or confirming fraud, using a combination of text analytics, statistical modelling, and classification.

CERTIFICATIONS

1. AWS Certified Data Engineer
2. AWS Certified Cloud Practitioner
3. AWS Certified AI Practitioner
4. AWS Certified DevOps Engineer - Professional
5. Microsoft Certified: Power BI Data Analyst Associate (PL-300)
6. Microsoft Certified: Azure Fundamentals (AZ-900)
7. Microsoft Certified: Azure Data Scientist Associate (DP-100)
8. Google Cloud Certified Professional Data Engineer
9. Google Cloud Certified Professional Cloud DevOps Engineer

Education:

Bachelor's in EE, Aurora's Technological and Research Institute, Hyderabad, India - 2016